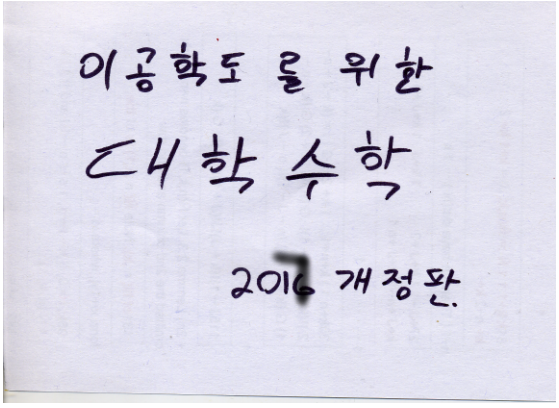
Calculus II Text Questions

These questions are from our text:



They are the exact questions, but translated (perhaps poorly) into (very good) English. You may work from the text, but you should do your questions from here to be sure you can understand the English. *The tests will be in English.*

Remember: you do not have to do any questions, but they will be on the quizzes and the tests.

Please let me know if you think I have mistranslated any questions or Section titles.

If you are a native English speaker, maybe work from the book instead.

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Chapter 7

Vector Functions

1	2
3	4

7.1 Vectors

1. For the given vectors $\mathbf{a}$ and $\mathbf{b}$, find $|\mathbf{a}|$, $\mathbf{a}+\mathbf{b}$ and $3\mathbf{a}+4\mathbf{b}$.

   1 $\mathbf{a} = \langle 4, 3, -1\rangle$, $\mathbf{b} = \langle 6, 2, 0\rangle$

   3 $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j}, \mathbf{b} = \mathbf{i} + 5\mathbf{j}$

2. (3) Find the unit vector in the same direction as $\mathbf{a} = 12\mathbf{i} - 5\mathbf{j}$.

3. Find a line that satisfies the following conditions.

   (1) Is parallel to the vector $\langle 2, -4, 5\rangle$ and contains the point $(1, 0, -3)$.

   (2) Goes through the points $(1, 3, 2)$ and $(-4, 3, 0)$.

   $\boxed{\langle x, y, z\rangle = \langle 1 - 5t, 3, 2 - 2t\rangle.}$

4. Show that if $k\mathbf{a} = 0$ for a vector $\mathbf{a}$ and a scalar k then either $k = 0$ or $\mathbf{a} = \mathbf{0}$.

7. (4) Find the angle between the vectors $\langle -1, 2, 5\rangle$ and $\langle 3, 4, -1\rangle$.

8. (1) Find b so that the vector $\langle 6, -b, 2\rangle$ and $\langle b, b^2, b\rangle$ are perpendicular.

9. (1) Find the projection $\text{proj}_{\mathbf{a}}\,\mathbf{b}$ of $\mathbf{b} = \langle 5, 0\rangle$ onto $\mathbf{a} = \langle 3, -4\rangle$.

11. Using inner products, prove

    (1) Cauchy-Schwartz: $|\mathbf{a}\cdot\mathbf{b}| \le |\mathbf{a}||\mathbf{b}|$.

    (2) $|\mathbf{a}+\mathbf{b}|^2 + |\mathbf{a}-\mathbf{b}|^2 = 2(|\mathbf{a}|^2 + |\mathbf{b}|^2)$.

13. (3) Find a unit vector normal to the plane containing the triangle $P(2, 0, -3)$, $Q(3, 1, 0)$, $R(5, 2, 2)$; and, find the area of the triangle.

15. (1) Find the volume of the parallelopiped defined by the line segments PQ, PR and PS for $P(2, 0, -1), Q(4, 1, 0), R(3, -1, 1)$ and $S(2, -2, 2)$.

17. (3) Find the equation of the plane that contains the point $(-1, 2, 1)$ and the intersection of the planes $x + y - z = 2$ and $2x - y + 3z = 1$.

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7.2 Vector Functions

1. (3) Determine the limit function of $\lim_{t\to 0}\langle \tan^{-1} t, e^{-2t}, \frac{\ln t}{t}\rangle$.
2. (1) Express the intersection of the surfaces $x^2 + y^2 = 1$ and $z = xy$ with a vector equation.
3. (3) Find the derivative of $\mathbf{r}(t) = \langle e^{t^2}, -1, \ln(1 + 3t)\rangle$.
4. Find the tangent vector of $\mathbf{r}(t) = \langle 6t^5, 4t^3, 2t\rangle$ at $t = 1$.
7. (1) Find the anti-derivative $\int \left(t^2\mathbf{i} - (3t - 1)\mathbf{j} - \frac{1}{t^3}\mathbf{k}\right) \mathrm{d}t$.
8. (3) Evaluate the anti-derivative $\int_0^1 \left(e^t\mathbf{i} + e^{-t}\mathbf{j} + 2t\mathbf{k}\right) \mathrm{d}t$.

7.3 Arc Length and Curvature

1. (1) Determine the length of the curve $\mathbf{r}(t) = \langle 2\sin t, 5t, 2\cos t\rangle$ for $-10 \le t \le 10$.
2. (3) Parametrise the curve $\mathbf{r}(t) = \langle 3\sin t, 4t, 3\cos t\rangle$ with respect to the length s of the curve (in the positive t direction) from $t = 0$.
3. (1) Find the curvature of the curve $\mathbf{r}(t) = t^2\mathbf{i} + t\mathbf{k}$.
5. (1) Find the curvature of the curve $y = x^4$.
6. (1) Find the vectors $\mathbf{T}, \mathbf{N}$ and $\mathbf{B}$ for $\mathbf{r}(t) = \langle t, t^2, \frac{2}{3}t^3\rangle$ at the point $(1, 1, \frac{2}{3})$.
7. Find the point at which the curvature of $y = e^x$ is maximum.
8. Find a point P on the curve C of $x = t^3, y = 3t, z = t^4$ for which the normal plane to C at P is parallel to the plane defined by $6x+6y-8z = 1$.

   $(-1, 3, 1)$, that is $t = -1$.

9. (1) Find the equations for the normal and osculating planes of the curve $\mathbf{r}(t) = \langle t, t^2, t^3\rangle$ at $t = 1$.

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Chapter 8

Partial Derivatives

8.1 Bivariate functions

1. (5) Determine the domain of the function $f(x, y) = \sqrt{1 - y^2} - \sqrt{1 - x^2}$.
2. (3) Draw some level curves for the functino $f(x, y) = x - y^2$.
3. (4) Draw the graph of $f(x, y) = \sqrt{x^2 + y^2}$.

8.2 Limits and Continuity

1. (1) Find $\lim_{(x,y)\to(0,0} x^2y + xy^2 + xy$.
2. Determine (and show) whether or not the following limits exist.

   (1) $\lim_{(x,y)\to(1,1} \frac{xy}{x^2+y^2}$

   (3) $\lim_{(x,y)\to(0,0} \frac{xy}{x^2+y^2}$
3. (3) Determine whether or not the function

$$f(x, y) = \begin{cases} \frac{x^3+y^3}{x^2+y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

   is continuous at $(0, 0)$.
4. (3) Determine where the function $f(x, y) = \ln |x^2 + y^2|$ is continuous.

8.3 Partial Derivatives

1. Find $f_x(x, y)$ and $f_y(x, y)$ for the following functions.

   (1) $f(x, y) = x^2y^3 + 2$.

   (7) $f(x, y) = x \ln(x^2 + y^2)$.
2. (3) Find f_{xy} and f_{yx} when $f(x, y) = \tan^{-1}\left(\frac{y}{x}\right)$.

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3. (1) Find the tangent plane to the graph of $f(x, y) = x - y^2$ at the point $P(1, 0, 1)$.

4. Show that for $f(x, y) = \frac{1}{e^{ay}} \cos ax$ the following is true:

$$f_{xx}(x, y) = a f_y(x, y).$$

6. Consider the following equations which describe the transformation from a cartesian coordinate system to a polar coordinate system (and back).

$$x = r\cos\theta \qquad y = r\sin\theta \qquad r = \sqrt{x^2 + y^2} \qquad \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

Find the following:

(1) $\frac{\partial x}{\partial r}$

(4) $\frac{\partial y}{\partial \theta}$

(7) $\frac{\partial \theta}{\partial x}$

7. (1) Show that the function $f(x, y) = e^x \sin y + e^y \cos x$ satisfies the following differential equation, known as Laplace's equation:

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$$

8.4 The Chain Rule

1. (3) Find the deriviative $\frac{\mathrm{d}z}{\mathrm{d}t}$ when $x = \sin xy + \cos x$, $x = t^2$, and $y = t$.

2. (3) Find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$ when $z = \ln(x^2 - y^2)$, $x = u - v$ and $y = u + v$.

3. (1) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ when $x^3 + 4x^2y - 2xy^2 + 3y^3 - 4 = 0$.

5. Show that the partial derivative of $f(x, y) = \sqrt{x^2 + y^2}$ with respect to x does not exist at $(0, 0)$.

7. Show that $\frac{\partial z}{\partial x} = -\frac{\partial z}{\partial y}$ holds when $z = f(x - y, y - x)$ for some function f.

10. Show that when all partial derivatives of $u = f(x, y)$ exist, and $x = e^s \cos t$ and $y = e^s \sin t$, then

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = e^{-2s}\left[\left(\frac{\partial u}{\partial s}\right)^2 + \left(\frac{\partial u}{\partial t}\right)^2\right]$$

8.5 Directional Derivatives

2. (3) Find the directional derivative of $f(x, y) = x\ln(x + y)$ at the point $P(-2, 3)$ in the direction of $\mathbf{v} = \langle -2\mathbf{i} + 3\mathbf{j}\rangle$.

3. (1) Find the directional derivative of $f(x, y) = x^2y^3 - y^4$ at the point $P(2, 1)$ in the direction of the vector that makes an angle of $\pi/4$ with the x axis.

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4. Find the direction and magnitude of the maximum direction derivative of $f(x, y) = \sqrt{xy}$ at the point $Q(5, 4)$.

6. The temperature of the point (x, y, z) on a surface is

$$T(x, t, z) = \frac{80}{1 + x^2 + 2y^2 + 3z^2}$$

degrees. Moving from the point $(1, 1, -2)$ in what direction is the temperature incresing the fastest?

7. (1) Find the equation of the tangent plane of surface $x^2 2y^2 + 3z^2 = 21$ at the point $(4, -1, 1)$, and of the normal line of the tangent plane.

8. Show that the symmetric form of the tangent plane to the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

at the point (x_0, y_0, z_0) is

$$\frac{xx_0}{a^2} + \frac{yy_0}{b^2} + \frac{zz_0}{c^2} = 1.$$

8.6 Minimums and Maximums

1. Find all local extrema and saddle points of

   (1) $f(x, y) = x^2 - 2y^2 - 6x - 7y + 2$

   (5) $f(x, y) = \cos x + \cos y$

2. Find the absolute minimum and maximum values of the given functions on the given domains.

   (3) $f(x, y) = x^2 + y^2 + xy^2 + 1$ on $D = \{(x, y) \mid |x| \leq 1, |y| \leq 1\}$.

   (5) $f(x, y) = 4x + 6y - x^2 - y^2$ on $D = \{(x, y) \mid 0 \leq x \leq 4, 0 \leq y \leq 5\}$.

3. (I couldn't understand this one, look at the book and translate it to English.)

4. Find the shortest distance between the point $(-1, 2, 1)$ and the plane $x + y - z = 2$.

5. Find the points on $z = xy + 2$ closest to the origin.

7. Find the minimum and maximum values of the function $f(x, y) = 4x^2 - 3y^2 + xy$ on the (closed) triangle defined by the points $(0, 0), (1, 0)$, and $(0, 1)$.

8. Of the sets of three positive numbers whose sum is 100, find one whose product is maximum.

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Chapter 9

Integrals

9.1 Integrals of two variables

1. Evaluate the following iterated integrals

   (3) $\int_1^3 \int_0^1 (1+2y)\,\mathrm{d}y\,\mathrm{d}x$

   (7) $\int_1^3 \int_0^1 x\sqrt{x^2+y}\,\mathrm{d}x\,\mathrm{d}y$

2. Evaluate the following integrals

   (3) $\int\int_R \frac{x}{1+xy}\,\mathrm{d}A$ where $R = [0,1]\times[0,1]$

   (7) $\int\int_R y^3\,\mathrm{d}A$ where R is the triangular region with corners $(0,2),(1,1)$ and $(3,2)$.

3. Evaluate $\int_0^8 \int_{\sqrt[3]{y}}^2 e^{x^4}\,\mathrm{d}x\,\mathrm{d}y$.

5. Using an integral in two variables find the volume of the region...

   (3) in the triangular cylindar through the triangle $(1,1),(4,1),(1,2)$ and between the surfaces $= 0$ and $z = xy$.

   (5) contained by the cylinder $x^2+y^2 = 1$ and the planes $z = 1-y$, $x = 0$ and $z = 0$.

9.2 Integrals using polar coordinates

1. Evaluate the following integrals.

   (3) $\int_{-3}^3 \int_0^{\sqrt{9-x^2}} \sin(x^2+y^2)\,\mathrm{d}y\,\mathrm{d}x$.

   (7) $\int_0^a \int_0^{\sqrt{a^2-x^2}} \frac{1}{(1+x^2+y^2)^{3/2}}\,\mathrm{d}y\,\mathrm{d}x$ (where $a > 0$).

2. Evaluate:

   (1) $\int\int_R e^{-(x^2+y^2)}\,\mathrm{d}A$ where R is bounded by $x = \sqrt{4-y^2}$ rotated around the y axis.

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(3) $\int\int_R x^2 \, \mathrm{d}A$ where R is the region outside the circle of radius 1 centered at the origin, and inside the circle of radius 3 centered at the origin.

(4) $\int\int_R x \, \mathrm{d}A$ where R is the region bounded between the ovals $x^2+y^2 = 4$ and $x^2+y^2 = 2x$.

3. Find the area of the region inside the graph of $r = 4\sin\theta$ ($0 \le \theta \le 2\pi$) and outside of $r = 2$.

5. Using integrals of two variables, find the volume of the following region.

(3) The region enclosed by the $z^2 - x^2 - z^2 = 1$ and $z = 2$.

(5) The region enclosed by the parabaloid $z = \sqrt{x^2+y^2}$ and the sphere $x^2+y^2+z^2 = 1$.

6. (1) Using the transformation $x = 2u$ and $y = 3v$ evaluate the integral $\int\int_R x^2 \, \mathrm{d}A$ over the region $R = \{(x,y) \mid 9x^2+4y^2 \le 36\}$.

9.3 Integrals in three variables

1. Evaluate the following integrals.

(1) $\int_0^1 \int_0^2 \int_{-1}^1 (2x+yz) \, \mathrm{d}x \, \mathrm{d}y \, \mathrm{d}z$

(7) $\int_0^{2\pi} \int_0^{\pi/4} \int_0^{2\sec\phi} \rho^2 \sin\phi \, \mathrm{d}\rho \, \mathrm{d}\phi \, \mathrm{d}\theta$

2. Reorder the integral $\int_{-1}^1 \int_{x^2}^1 \int_0^{1-y} f(x,y,z) \, \mathrm{d}z \, \mathrm{d}y \, \mathrm{d}x$ so that it is an integral over $\mathrm{d}x \, \mathrm{d}y \, \mathrm{d}z$.

3. What is the point $(r,\theta,z) = (\sqrt{2}, \frac{3\pi}{4}, 2)$ in rectilinear (x,y,z) coordinates.

4. What is the point $(\rho,\theta,\phi) = (4, -\frac{\pi}{4}, \frac{\pi}{3})$ in rectilinear (x,y,z) coordinates.

5. Write $(x,y,z) = (1,0,\sqrt{3})$ in spherical and cylindrical coordinates.

7. Using cylindrical or spherical coordinates, evaluate the following integrals.

(1) $$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{x^2+y^2}^{2-x^2-y^2} (x^2+y^2)^{3/2} \, \mathrm{d}z \, \mathrm{d}y \, \mathrm{d}x$$

(3) $$\int_{-3}^3 \int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} \int_{-\sqrt{9-x^2-y^2}}^{\sqrt{9-x^2-y^2}} \sqrt{x^2+y^2+z^2} \, \mathrm{d}z \, \mathrm{d}x \, \mathrm{d}y$$

8. Do these.

(1) $\int\int\int_G (xz - y^3) \, \mathrm{d}V$ where

$$G = \{(x,y,z) \mid -1 \le x \le 1, 0 \le y \le 2, 0 \le z \le 1\}.$$

(7) $\int\int\int_G \cos(z/y) \, \mathrm{d}V$ where

$$G = \{(x,y,z) \mid y \le x \le \pi/2, \pi/6 \le y \le \pi/2, 0 \le z \le xy\}.$$

(9) $\int\int\int_G z \, \mathrm{d}V$ where G is contained between $z = x^2+y^2$ and $z = 4$.

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(13) $\int\int\int_G y^2 \, \mathrm{d}V$ where G is the set of points satisfying $x^2 + y^2 + z^2 \leq 9$ and $y \geq 0$.

9. Using a triple integral, find the volume of the tetrahedron enclosed by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$ and $z = 0$.

11. Find the volume of the region enclosed between the paraboloid $x = y^2 + z^2$ and the plane $x = 16$.

13. Using spherical coordinates, find the volume of the region inside the sphere $x^2 + y^2 + z^2 = 4$ and between the planes $z = 0$ and $z = 1$.

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Chapter 10

Vector Analysis

10.1 Line Integrals

1. Evaluate the following line integrals

   (3) $\int_C xy^4 \, \mathrm{d}s$ where C is the right half of the circle $x^2 + y^2 = 1$.

   (7) $\int_C xy \, \mathrm{d}x + (x - y) \, \mathrm{d}y$ where C is the curve starting at $(0, 0)$ going in a straight line to $(2, 0)$ and then from there going in a straight line to $(3, 2)$.

   (11) $\int_C (x + yz) \, \mathrm{d}x + 2x \, \mathrm{d}y + xyz \, \mathrm{d}z$ where C is the piecewise linear curve from $(1, 0, 1)$ to $(2, 3, 1)$ to $(2, 5, 2)$.

2. Evaluate $\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}$ where...

   (1) $\mathbf{F}(x, y) = \langle -x^2y^3, -y\sqrt{x} \rangle, \mathbf{r}(t) = \langle t^2, t^3 \rangle$ for $0 \leq t \leq 1$.

   (5) $\mathbf{F}(x, y) = \langle x, -z, y \rangle, \mathbf{r}(t) = \langle 2t, 3t, -t^2 \rangle$ for $-1 \leq t \leq 1$.

(3) Find the work done by a force of $\mathbf{F}(x, y, z) = x\mathbf{i} + (y + 2)\mathbf{j}$ moving a mass along the path $\mathbf{r}(t) = (t - \sin t)\mathbf{i} + (1 - \cos t)\mathbf{j}$ for $0 \leq t \leq 2\pi$.

10.2 Fundamental Theorem of Line Integrals, and Greens Theorem

1. Determine if the following vector fields are conservative, and if they are, find their potential functions.

   (1) $\mathbf{F}(x, y) = \langle 6x + 5y, 5x + 4y \rangle$

   (3) $\mathbf{F}(x, y) = e^y\mathbf{i} + xe^y\mathbf{j}$

   (5) $\mathbf{F}(x, y) = \langle y^2, (2xy + e^{3z}), 3ye^{3z} \rangle$

2. Evaluate $\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}$ given the following $\mathbf{F}$ and C.

   (1) $\mathbf{F}(x, y) = \langle y, x + 2y \rangle$, and C is the line segment from $(0, 1)$ to $(2, 1)$.

   (3) $\mathbf{F}(x, y, z) = (1 + xy)e^{xy}\mathbf{i} + x^2e^{xy}\mathbf{j}$, and C is $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$ for $0 \leq t \leq 1$.

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(5) $\mathbf{F}(x,y,z) = \langle yz, xz, xy+2z\rangle$ and C is the line from $(1,0,-2)$ to $(4,6,3)$.

3. Use Green's Theorem to evaluate $\int_C x^2y^2\,\mathrm{d}x + 4xy^3\,\mathrm{d}y$ where C is the (counterclockwise) boundary of the triangle with points $(0,0),(1,3)$ and $(0,3)$.

4. Use Green's Theorem to evaluate $\int_C \mathbf{F}\cdot\mathrm{d}\mathbf{r}$ where...

   (1) $\mathbf{F}(x,y) = \langle \sqrt{x}+y^3, x^2+\sqrt{y}\rangle$ and C is the curve that follows $y=\sin x$ from $(0,0)$ to $(\pi,0)$ and then takes the straight line back to $(0,0)$.

   (3) $\mathbf{F}(x,y) = \langle e^x+x^2y, e^y-xy^2\rangle$ and C is the clockwise cycle $x^2+y^2=25$.

5. Use Green's theorem to evaluate $\int_C x^2y\,\mathrm{d}x - y^2x\,\mathrm{d}y$ where C is the curve that goes counterclockwise around first quadrant of the circle $x^2+y^2=4$.

7. Use a line integral to find the area of the triangle with vertices $(0,0),(a,0)$ and $(0,b)$, where a and b are positive numbers.

9. Let C be the curve going once counterclockwise around the ellipse $\frac{x^2}{3}+\frac{y^2}{5}=1$. Let R be the region it bounds. Computing both $\int_C f(x,y)\,\mathrm{d}x + g(x,y)\,\mathrm{d}y$ and $\int\int_R \left(\frac{\partial g}{\partial x}-\frac{\partial f}{\partial y}\right)\mathrm{d}A$, for $f(x,y)=x^3y^2$ and $g(x,y)=xy^2$, verify that Green's Theorem holds.

11. Let $\mathbf{F} = \langle \frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2}\rangle$. Show that $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ but that $\int_C \mathbf{F}\cdot\mathbf{r}$ is not 0 when C goes once around the unit circle around the origin. Explain why the fundamental theorem of line integrals does not hold.

10.3 Surface Integrals

1. (3) Using spherical coordinates describe the boundary of the part of the sphere $x^2+y^2+z^2=4$ that lies between the planes $z=-2$ and $z=2$.

2. Give the equation for the tangent plane to the surface $r(u,v) = \langle u^2+1, v^3+1, u+v\rangle$ at the point $(5,2,3)$.

3. Find the area of the part of the plane $x+2y+z=4$ contained in the cylinder $x^2+y^2=4$.

4. Evaluation the following surface integrals

   (1) $\int\int_S x^2yz\,\mathrm{d}S$ where S is the part of the plane $z=1+2x+3y$ over the rectangle $[0,3]\times[0,2]$ in the xy-plane.

   (5) $\int\int_S xyz\,\mathrm{d}S$ where S is the part of the sphere $x^2+y^2+z^2=1$ above $z=\sqrt{x^2+y^2}$.

5. Find the flux of the given field $\mathbf{F}$ through the given surface S.

   (1) $\mathbf{F}(x,y,z) = \langle xy, yz, zx\rangle$ and S is the part of the paraboloid $z=4-x^2-y^2$ above the square $0\le x,y\le 1$ (oriented up).

   (3) $\mathbf{F}(x,y,z) = \langle x,-z,y\rangle$ and S is the portion of the sphere $x^2+y^2+z^2=4$ in the first octant.

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10.4 Stoke's theorem and the Divergence Theorem

1. Give the curl and the divergence of $\mathbf{F}(x, y, z) = \langle e^x \sin y, e^x \cos y, x\rangle$.

2. Where f is a scalar function, and $\mathbf{F}$ and $\mathbf{G}$ are vector fields, show that $\operatorname{div}(\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot \operatorname{curl} \mathbf{F} - \mathbf{F} \cdot \operatorname{curl} \mathbf{G}$.

3. (1) Use Stoke's theorem to evaluate $\int\int_s \operatorname{curl} \mathbf{F} \cdot d\mathbf{S}$, when $\mathbf{F}(x, y, z) = \langle yz, xz, xy\rangle$ and S bounds the portion of $z \leq 9 - x^2 - y^2$ above $z = 5$.

4. (3) Use Stoke's theorem to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F}(x, y, z) = yz\mathbf{i} + 2xz\mathbf{j} + e^{xy}\mathbf{k}$ and C (traversed counterclockwise) is the circle with $z = 5$ and $x^2 + y^2 = 16$.

5. Use the divergence theorem to calculate the flux $\int\int_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F}$ through S (outwardly oriented) when:

   (1) $\mathbf{F}(x, y, z) = \langle e^x \sin y, e^x \cos y, yz^2\rangle$ and S is the surface of the rectangular prism bounded by $x = 0, x = 1, y = 0, y = 1, z = 0$ and $z = 2$

   (3) $\mathbf{F}(x, y, z) = \langle x^2y, xy^2, 2xyz\rangle$ and S is the surface of the region bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $x + 2y + z = 2$.

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